# Labamu Manufacturing User Guide

| | |
|---|---|
| **Document Owner** | @Devin Jastona |
| **Product** | @Ivan Syafarwan @Devin Jastona @Gabriella Anggita @Sekarwulan @Hana Hanifah |
| **Tech** | @Naomi Olga Panjaitan @Maulana Akbar Yudistika @Zikry Kurniawan @Muhammad Anshor |
| **QA** | |
| **Stakeholders** | @Emmanuel Van De Geer @Kamal Shivdasani @Silvya Winsen |

## User Guide

Complete step-by-step guide to using **Labamu Manufacturing (MRP)**

This guide walks you through the entire manufacturing workflow, from setup to production tracking. You can plan, produce, and monitor efficiently.

---

#### Step 1: Getting Started

Set up your account and understand the basics.

**What you need to know:**

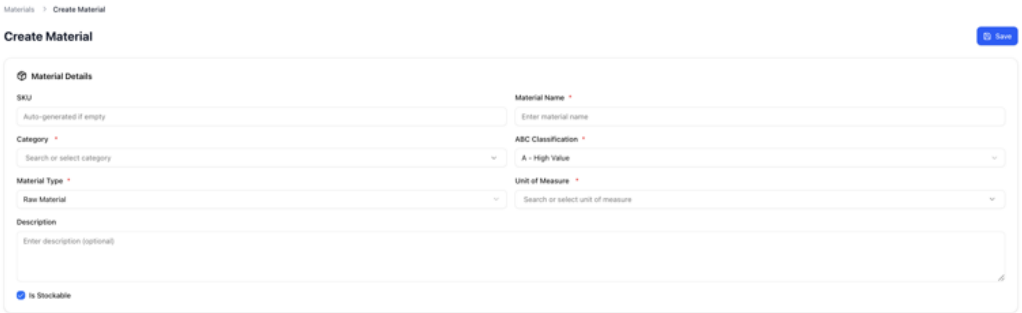
- MRP is automatically available once your business is activated for the Manufacturing app on Labamu.
- It connects directly with **Products**, **Materials**, **BOM (Bill of Materials)**, and **Routing**.
- Each production order follows this flow:
  **Material Planning → RFQ → Quotes → Order → Work Order → Production Tracking → Completion**
- Ensure your master data (Products, Materials, BOM, and Routing) is uploaded before creating an Order.

---

**Step 2: Manage Materials**

Define all raw materials or components used in your production.

**How to Create Materials:**

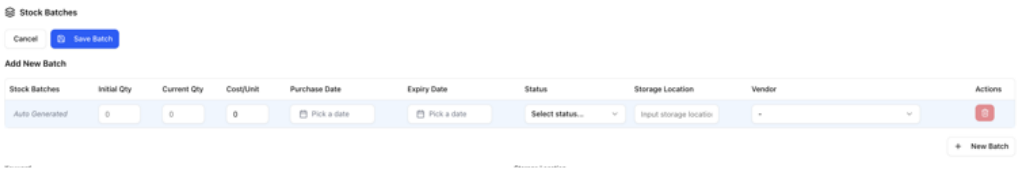



1. Navigate to **Manufacturing → Materials** in the sidebar.
2. Click **New Material**.
3. Fill in the details as shown below:
   - **SKU** – Auto-generated if left empty.
   - **Material Name** – (Mandatory) Enter the material name (e.g., *Steel Plate 2mm*).
   - **Category** – (Mandatory) Select or create a material category (e.g., *Metal*, *Plastic*).
   - **ABC Classification** – (Mandatory) Choose classification based on value or importance (A = High Value, B = Medium, C = Low).
   - **Material Type** – (Mandatory) Choose *Raw Material*, *Sub Material*, or *Consumable*.
   - **Unit of Measure** – (Mandatory) Select or create the measurement unit (e.g., *pcs*, *kg*, *mL*).
   - **Description** – (Optional) Add a short note about the material.
   - **Is Stockable** – (Optional) Keep this checked if the material should be tracked in inventory.
   - **Image** – (Optional) Upload an image to help visually identify the material.
4. Click **Save** at the top right corner.

**Tips:**

- **SKU** will be automatically generated if left blank.
- You can use **Bulk Upload** to import multiple materials via Excel template.
- Only **Active** materials can be used in **BOM** (Bill of Materials) and **Work Orders**.
- Maintain consistent naming conventions for easier tracking and reporting.

**How to Create Batches:**




Each material can have multiple **batches** to track stock quantity, expiry, and vendor details. This helps you manage materials more accurately.

1. Open the **material** you want to manage (for example, *Table*).
2. Go to the **Stock Batches** tab.
3. Click **+ New Batch**.
4. Fill in the following details:
   - **Initial Quantity** — (Mandatory) Enter the total quantity received or produced (e.g., *100*).
   - **Cost per Unit** — (Optional) Input the cost of the material per unit. (e.g., *100.000*)
   - **Purchase Date** — (Mandatory) Select the date when the material was received or purchased.
   - **Expiry Date** — (Optional) Add an expiry or best-before date if applicable.
   - **Status** — (Mandatory) Choose one of the following:
     - *Requested* – Batch requested but not yet received.
     - *Expected* – Batch is expected to arrive.
     - *Delayed* – Delivery delayed.
     - *Received* – Material has been received and available in stock.
   - **Storage Location** — (Optional) Enter where the batch is stored (e.g., *Warehouse A, Rack 3B*).
   - **Vendor** — (Optional) Select the vendor if applicable.
5. After filling all fields, click **Save Batch**.
6. To add another batch, click **+ New Batch** again.

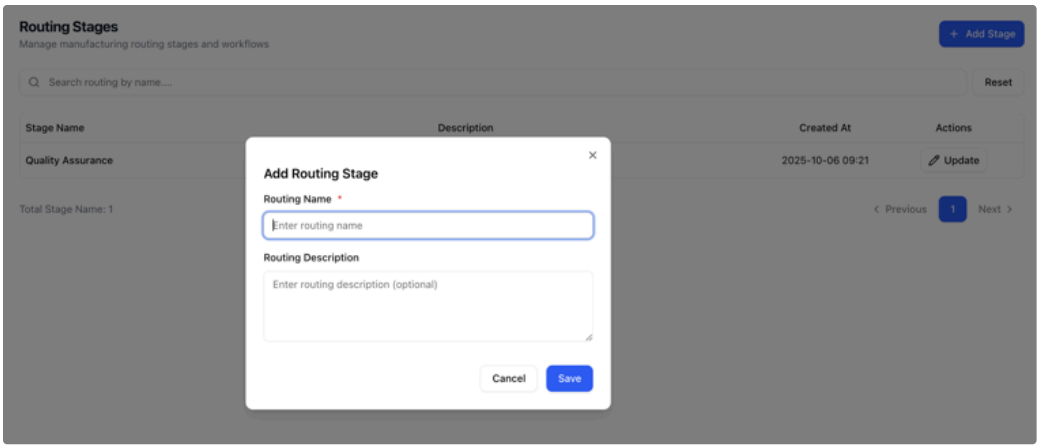
**Notes:**

- The system automatically generates a **Batch ID** — no need to fill it manually.
- **Current Quantity** will follow the **Initial Quantity** entered during the first creation of the batch.
- You can filter batches by **Status**, **Vendor**, or **Expiry** using the search options below the batch list.
- Use **Show Empty Batches** to display all inactive or depleted batches.
- Always ensure **purchase and expiry dates** are valid to avoid errors.

#### Step 3: Create Routing

Define the sequence of stages required to manufacture a product. Each routing stage represents a specific operation in your production process.

#### How to Create Routing




1. Navigate to Manufacturing → Routing in the sidebar.
2. Click Add Stage on the top right corner.
3. Fill in the following details in the popup form:
   - Routing Name – (Mandatory) Enter the stage name (e.g., Cutting, Assembly, Quality Assurance).
   - Routing Description – (Optional) Add details about the operation or purpose of this stage.
4. Click Save to create the routing stage.

**Notes:**

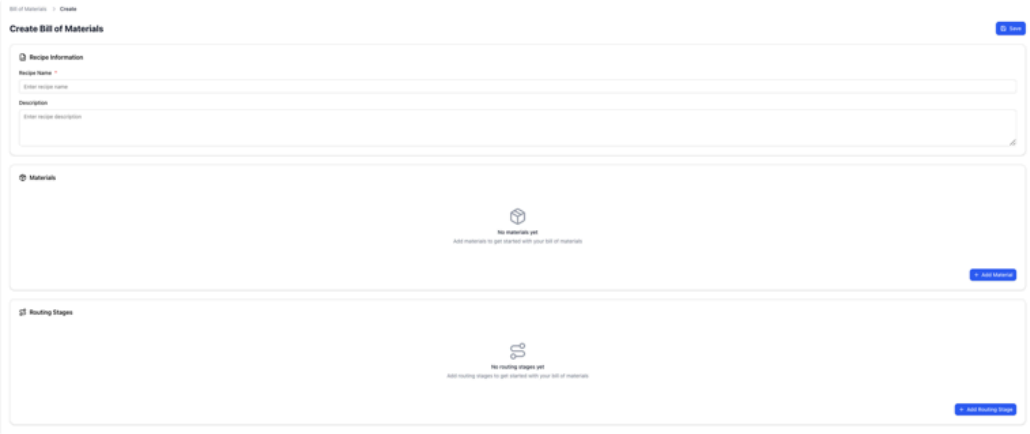
- You can create multiple routing stages to define your full production workflow.
- Keep stage names clear and consistent (e.g., Cutting → Welding → Painting → Quality Check).
- Routing stages will be used later in Work Orders to define the manufacturing sequence and progress tracking.

---

#### Step 4: Create Bill of Materials (BOM)

Define the list of materials and production stages required to manufacture a product. The Bill of Materials (BOM) connects materials, quantities, and routing steps into a complete production recipe.

**How to Create a BOM**




1. Navigate to Manufacturing → Bill of Materials in the sidebar.
2. Click New BOM at the top right corner.
3. Fill in the Recipe Information:
   - Recipe Name – (Mandatory) Enter a unique name for this recipe (e.g., *Recipe A*).
   - Description – (Optional) Add notes about the purpose or version of the BOM.
4. Add Materials:
   - Click Add Material.
   - Select a Material from the dropdown (Mandatory).
   - Enter the Quantity required per unit of product (Mandatory) (e.g., *2*).
   - Click Add Material to confirm.
5. Add Routing Stages (optional):
   - Click Add Routing Stage.
   - Select a Stage Name (Mandatory) (e.g., *Cutting Raw Material*)
   - Choose the Routing workflow.
   - Enter the Step Number (sequence order) and estimated Hours (Mandatory) (e.g., *2 hours*).
   - Click Add Routing Stage to confirm.
6. Add Forecasted Cost of Goods Sold:
   - **Material Cost** – Automatically calculated based on the average material price available.
   - **Labour Cost** *(Optional)* – Cost of labor required to produce one unit.
   - **Packing Cost** *(Optional)* – Cost of packaging the product.
   - **Other Cost** *(Optional)* – Any additional costs.
7. When all details are complete, click Save at the top right corner.

**Notes:**

- You can mark a BOM as Active or Non-Active.

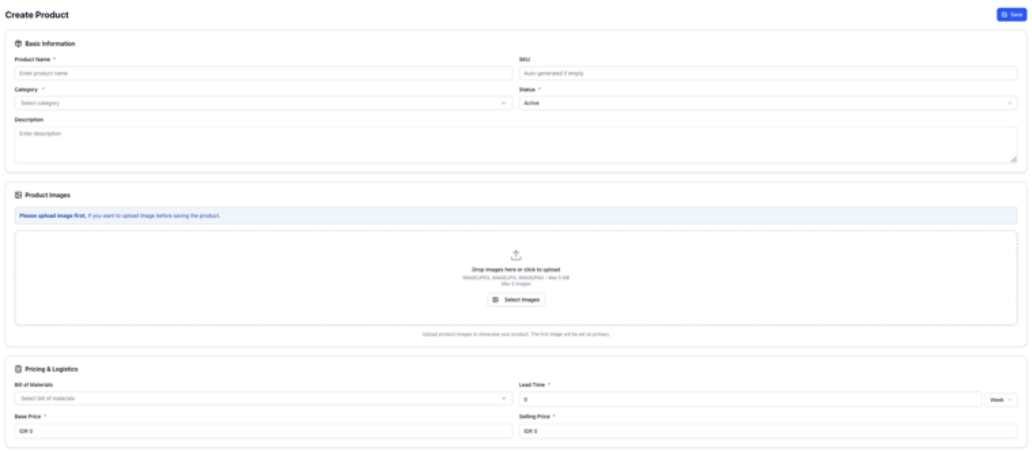
- Changing a BOM to Non-Active will automatically unlink it from all associated products.
- Only Active BOMs can be used in Work Orders and Production Scheduling.
- The **total Forecasted Cost of Goods Sold** will be used as the **Base Price** of the product when selecting a BOM in the product setup.

---

#### Step 5: Create Products

Define finished goods or semi-finished products that will be produced using materials and routing. Each product serves as the final output of your Bill of Materials (BOM).

#### How to Create a Product




1. Navigate to Products → Product in the sidebar.
2. Click New Product at the top right corner.
3. Fill in the Basic Information section:
   - Product Name – (Mandatory) Enter the product name (e.g., Stainless Steel Table).
   - SKU – (Optional) Automatically generated if left empty (e.g., TBL-002).
   - Category – (Mandatory) Select or create a product category (e.g., Furniture, Electronics).
   - Status – (Mandatory) Choose Active if the product should be available for use (e.g., Active).
   - Description – (Optional) Add additional details about the product.
4. (Optional) Upload Product Images:
   - Click Select Images or drag and drop image files.
   - The first uploaded image will be set as the primary display image.
5. In the Pricing & Logistics section:
   - Bill of Materials – (Optional) Select the BOM that defines how this product is made.
   - Base Price – (Mandatory) Input the cost per unit (e.g., IDR 100.000).

- Selling Price – (Mandatory) Enter the final price that will appear in sales modules (e.g., IDR 200.000).
- Lead Time – (Mandatory)  Define the estimated production time (e.g., 2 weeks).

6. Click Save at the top right corner to create the product.

**Notes:**

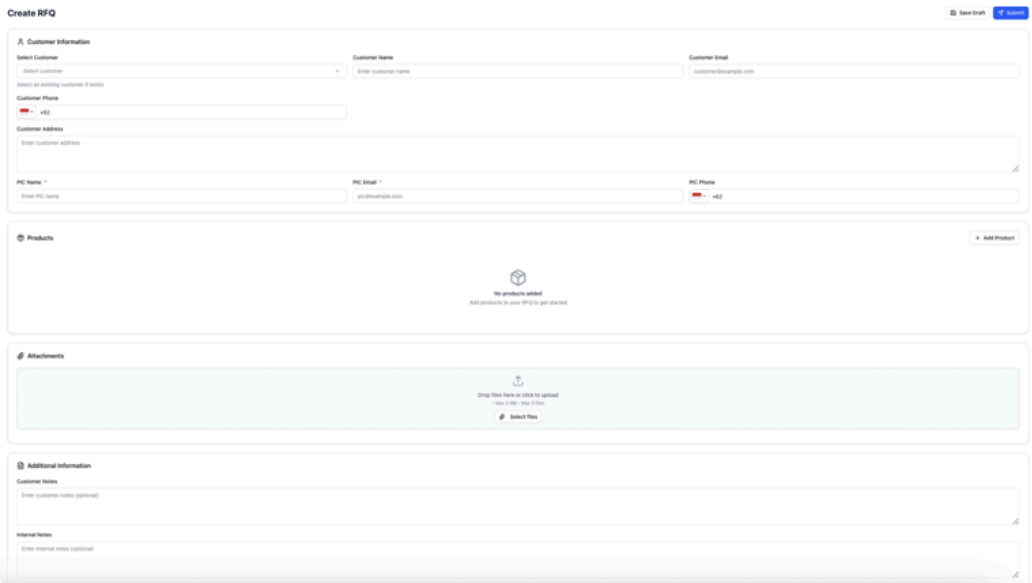
- Products can be linked directly to a BOM and Routing to define how they're produced.
- Keep product names clear and consistent with BOM naming conventions.
- You can edit or deactivate products anytime from the product list.
- Uploading product images helps for catalog or sales purposes.
- Product that created without BOM cannot be use for Creating RFQ, Quote, Order
- Product Inactive cannot be use for Creating RFQ, Quote, Order

---

**Step 6: Request for Quotation (RFQ)**

Plan and manage your material procurement process by sending RFQs (Requests for Quotation) to suppliers based on your production needs.

**How to Create an RFQ**




1. Navigate to **Procurement → RFQ**.
2. Click **New RFQ** to open the creation form.
3. Fill in the **Customer Information** section:
   - **Select Customer** – (Optional) Choose an existing customer or left it blank if u want create new customer
   - **Customer Name** – (Mandatory) Fill in relevant customer name (e.g., XYZ Corp.).

- **Customer Email** – (Optional) Fill in relevant customer email (e.g., admin@xyz.com).
- **Customer Phone** – (Optional) Fill in relevant customer phone (e.g., +6287384748392).
- **Customer Address** – (Mandatory) Fill in relevant customer address (e.g., Street Wall No.24).
- Add Multiple PIC (Max 5 PIC). Each PIC includes the following fields:
  - **PIC Primary** – Flag to mark as the primary PIC (**only one** PIC should be primary).
  - **PIC Name** – (Mandatory) Enter the PIC's name (e.g., *John Doe*).
  - **PIC Email** – (Mandatory) Enter the PIC's email (e.g., *johndoe@mail.com*).
  - **PIC Phone** – (Mandatory) Enter the PIC's phone number (e.g., *+6287384748392*).
  - **PIC Role** – (Mandatory) Select *Viewer* or *Approver*. This determines actions allowed on the Microsite.

4. Under **Products**, click **Add Product**:
   - Select products.
   - In the **Add Product to Quote** popup:
     - **Product** – (Mandatory) Select or search for the product.
     - **Quantity** – (Mandatory)  Input the number of units (e.g., 2).
     - **Attachments** – (Optional) Upload up to 3 files (max 5 MB each).
5. (Optional) Upload relevant **Attachments** such as product specifications or technical drawings.
6. Add **Additional Information**:
   - **Customer Notes -** (Optional)
   - **Internal Notes** – (Optional) Notes for internal team reference only.
7. Once completed, choose one of the following actions:
   - **Save Draft** – To continue editing later.
   - **Submit** – To proceed for approval.

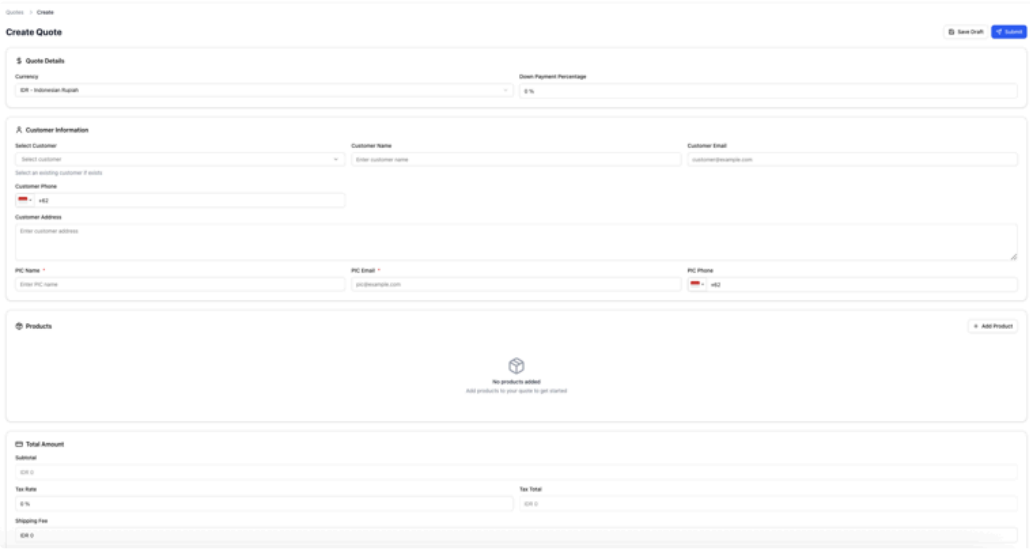
#### Approval Flow

- Submitted RFQs will follow the **Approval Rules** defined in the **RFQ Settings**.
- If no approval rule exists, the RFQ will be **auto-approved** and automatically move to the **Quotation** stage.
- If approval is required:
  - The RFQ will be routed to the designated approver.
  - If **approved**, it proceeds to **Quotation** creation.
  - If **rejected**, the RFQ cannot continue to the next stage.

---

#### Step 7: Manage Quotes

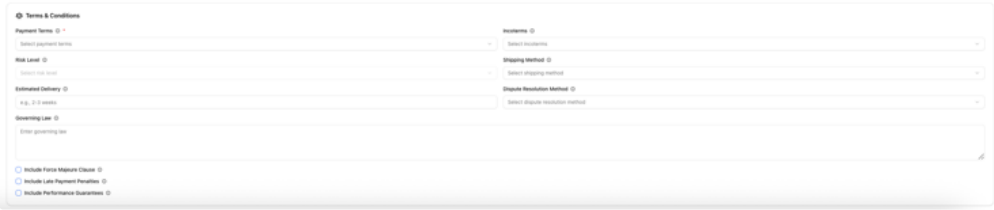
Create, manage, and compare customer quotations before confirming an order.




#### How to Create a Quote

1. **Navigate to** `Sales → Quotes` in the sidebar.
2. **Click** `Create Quote` on the top-right corner.
3. **Fill Quote Details:**
   - **Currency** – Select the transaction currency (e.g., IDR – Indonesian Rupiah).
   - **Down Payment Percentage** – Define the required down payment (optional, default 0%).
4. **Fill Customer Information:**
   - **Select Customer** – (Optional) Choose an existing customer or left it blank if u want create new customer
   - **Customer Name** – (Mandatory) Fill in relevant customer name (e.g., XYZ Corp.).
   - **Customer Email** – (Optional) Fill in relevant customer email (e.g., admin@xyz.com).
   - **Customer Phone** – (Optional) Fill in relevant customer phone (e.g., +6287384748392).
   - **Customer Address** – (Mandatory) Fill in relevant customer address (e.g., Street Wall No.24).
   - Add Multiple PIC (Max 5 PIC). Each PIC includes the following fields:
     - **PIC Primary** – Flag to mark as the primary PIC (**only one** PIC should be primary).
     - **PIC Name** – (Mandatory) Enter the PIC's name (e.g., *John Doe*).
     - **PIC Email** – (Mandatory) Enter the PIC's email (e.g., *johndoe@mail.com*).
     - **PIC Phone** – (Mandatory) Enter the PIC's phone number (e.g., *+6287384748392*).
     - **PIC Role** – (Mandatory) Select *Viewer* or *Approver*. This determines actions allowed on the Microsite.

5. **Add Products to the Quote:**
   - Click **+ Add Product**.
   - In the **Add Product to Quote** popup:
     - **Product** – Select or search for the product.
     - **Quantity** – (Mandatory) Input the number of units  (e.g., 2)..
     - **Unit Price** – (Mandatory) Enter the price per unit (e.g., IDR 200.000)..
     - **Discount** – (Optional) Add a discount if applicable (e.g., 0%).
     - **Total Price** – Automatically calculated based on quantity and discount  (e.g., IDR 200.000).
     - **Notes** – (Optional) Add internal or customer notes.
     - **Attachments** – (Mandatory) Upload up to 3 files (max 5 MB each), such as product specs documents.
6. **Total Amount:**
   After products are added, the system automatically calculates totals. Complete the following fields:
   - **Subtotal** – Total before taxes or fees (e.g., IDR 200.000)
   - **Tax Rate** – (Optional) Input tax rate (e.g., 10%).
   - **Tax Total** – Total updates automatically. (e.g., IDR 20.000)
   - **Shipping Fee** – (Optional) Optional shipping cost. (e.g., IDR 10.000)
   - **Other Fee** – (Optional) Add additional charges if any. (e.g., IDR 20.000)
   - **Total** – Final amount after tax and all fees. (e.g., IDR 250.000)
7. **Add Attachments (Optional):**
   Upload general files related to the quotation, such as proposals, photos, or supporting documents.
   You can upload up to 3 files (max 5 MB each).
8. **Fill Terms & Conditions:**




   Define agreement details before submission:

- **Payment Terms** – (Mandatory) Select available payment options (e.g., NET 30 – payment due in 30 days after invoice, NET 60, or Cash Before Delivery).
- **Incoterms** – (Optional) Select delivery terms if applicable (e.g., FOB – Free on Board, CIF – Cost, Insurance, and Freight, or EXW – Ex Works).
- **Risk Level** – Automatically filled with the project or transaction risk level based on the selected Incoterms (e.g., Medium Risk for international shipment using CIF terms).
- **Estimated Delivery** – (Optional) Input expected delivery time (e.g., 2–3 weeks after order confirmation).
- **Governing Law** – (Optional) Define applicable law or jurisdiction (e.g., Indonesian Law or Singapore Commercial Law).
- **Shipping Method** – (Optional) Choose the transportation method (e.g., Air Freight, Sea Cargo, or Local Courier).
- **Dispute Resolution Method** – (Optional) Define how disputes will be resolved (e.g., Arbitration, Mediation, or Court Settlement).
- **Include Clauses (Optional):**
  - **Include Force Majeure Clause** – (Optional) Protects both parties if unexpected events (e.g., disasters) prevent fulfillment of the agreement.
  - **Include Late Payment Penalties** – (Optional) Applies fees or interest if the customer pays past the due date.
  - **Include Performance Guarantees** – (Optional) Ensures the seller meets agreed quality or delivery standards.

9. **Final Actions:**
   After filling all required details:
   - **Save Draft** – Save for later editing.
   - **Submit** – Send the quotation for approval.

#### Notes:

- If you want to use a currency other than your base currency, please ensure your **FX Management** module is updated. You must input the **latest FX exchange rate** to ensure accurate conversion during quotation.

#### Quote Status Flow

- **Draft** – Quotation is being prepared and not yet submitted.
- **Submitted** – Waiting for internal approval (based on approval rules).
- **Issued** – Approved internally and sent to the customer.
- **Approved** – Customer has accepted the quotation.

- **Rejected** – Rejected by internal approver or customer.
- **Need Revision** – Customer requested updates or corrections before approval.

#### Approval Flow

- Internal approval follows the rules defined in **Quote Settings → Approval Rules**.
- Once internally approved, the quote changes to **Issued** and awaits customer approval.
- The customer must approve before the quote can be converted into an **Order**.
- If no approval rule exists, the system automatically moves from **Submitted → Issued**.

#### After Approval

- Once **Approved** by the customer:
  - The system automatically converts the quote into an **Order and Work Order**.
  - All pricing, product, and payment details are carried forward.
  - If Downpayment is filled, system will automatically create Downpayment Invoice.
- **Rejected** quotes cannot proceed to the next stage.

---

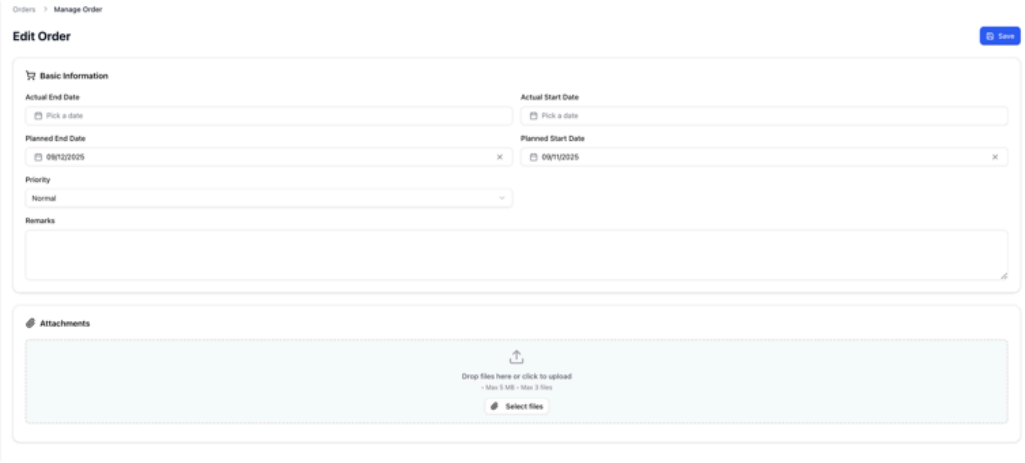
#### Step 8: Manage Orders

Confirm and manage customer orders to proceed with manufacturing and shipment. You can also update order details, attach documents, and track related invoices, work orders, and shipment records.

#### How to View Orders

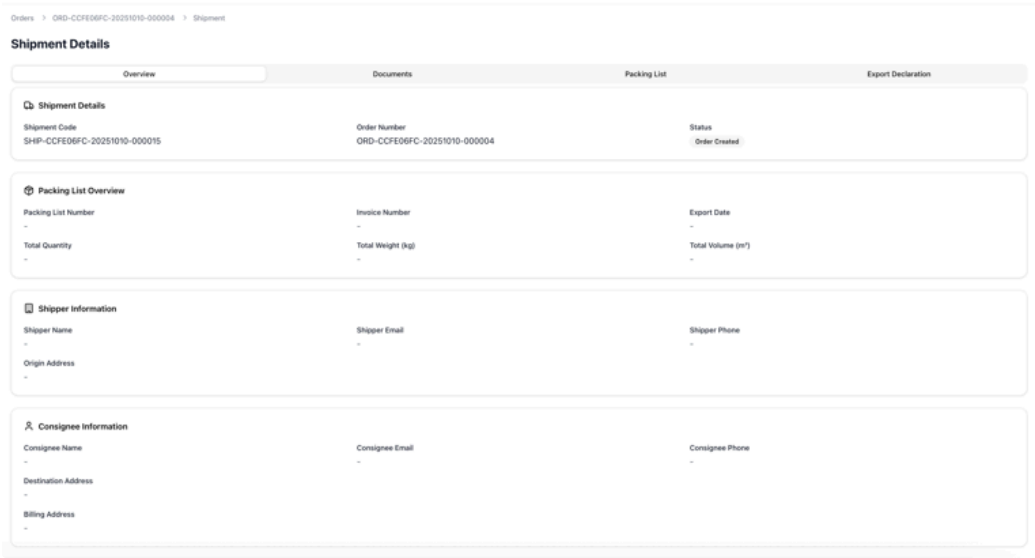
1. Navigate to Procurement → Orders.
2. Select the desired Order from the list to open the Order Details page.

#### How to Edit an Order



1. On the Order Details page, click Edit in the top-right corner.
2. You can update the following fields:
   - **Planned Start Date** – Input the target start date for the order (e.g., 15 Oct 2025).
   - **Planned End Date** – Input the estimated completion date for the order (e.g., 25 Oct 2025).
   - **Actual Start Date** – Record the actual date when the order process begins (e.g., 16 Oct 2025).
   - **Actual End Date** – Record the actual date when the order process is completed (e.g., 24 Oct 2025).
   - **Priority** – Define the priority level of the order (e.g., Low, Medium, High, or Urgent).
   - **Remarks** – Add any additional notes or important information related to the order (e.g., “Expedite delivery due to customer request”).
   - **Purchase Order No** – PO number provided by the buyer.
3. You can also add Attachments related to the order (e.g., additional files, supporting documents).
4. Click Save to confirm changes.

#### How to Manage Shipment




1. In the Order Details page, click on the Shipment Code.
2. The system will open the Shipment Details page.
3. You can view and manage shipment records for tracking and documentation purposes.

   Note: Shipment records here are for documentation and reference only.

#### Order Status Flow

- Not Started – Order has been created but not yet processed.

- Waiting for Approval – Pending internal approval based on approval rules.
- Confirmed – Order has been approved and is ready for manufacturing.
- In Progress – Work Order related to this Order is ongoing.
- Ready to Ship  – All Work Orders under this Order are finished and Ready to ship.
- On Shipping - On going Shipping
- Completed – All processes are complete, and the final invoice has been fully paid.
- Canceled – Order has been rejected or canceled.

#### Approval Flow

- Follows approval rules defined in Order Settings → Approval Rules.
- Once approved, the Order status changes from Waiting for Approval → Confirmed.
- If no approval rule exists, the Order will move directly from Not Started → Confirmed.

---

#### Step 9: Manage Work Order

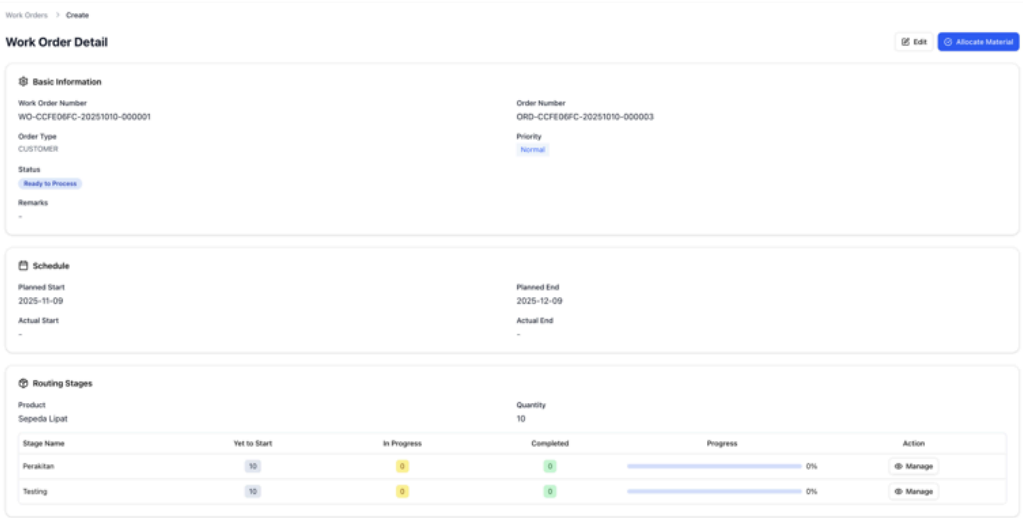
Start the production process based on your Bill of Materials (BOM) and Routing. Work Orders is automatically created once the Quote is approved by the customer along with the Order creation. Work Order can only be started when the Order is Confirmed and the materials are allocated.

#### How to View Work Orders

1. Navigate to Procurement → Work Orders.
2. Select the desired Work Order from the list to open the Work Order Detail page.

#### How to Allocate Materials

Before production starts, materials must be available in the inventory.



To allocate materials:

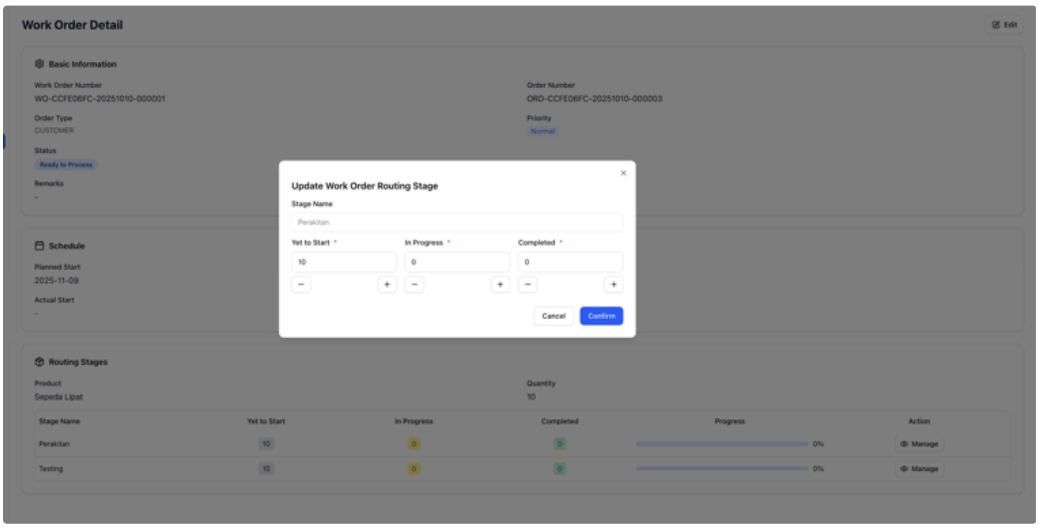
1. Open the desired Work Order.
2. Click Ready to Process (manual process from Order page) or it will be set automatically if the Order is already confirmed — this will display the Allocate Materials button.
3. Click Allocate Materials.
4. The system will show a confirmation:
   - If sufficient stock → the system allocates materials and shows a success message.
   - If insufficient stock → the system displays an error and cannot allocate.

Notes:

- You cannot start production without allocation.
- The Allocate Materials button only appears when the status is Ready to Process.

#### How to Manage Routing Stages

After successful material allocation, you can manage and track progress for each Routing Stage.




To update routing progress:

1. Go to the Routing Stages tab inside the Work Order.
2. For each stage, click Manage.
3. Update the quantities for Yet to Start, In Progress, or Completed.
4. Click Save Progress.
5. The system automatically calculates progress percentage.

Notes:
Routing stages can only be managed after successful material allocation.
Updating progress in any routing stage will automatically change the status to In Progress.
Alternatively, you can manually click the Start Process button to begin.

#### How to Complete Work Order

Once all routing stages are completed, mark the production as finished.

To complete:

1. Verify that all routing stages are 100% completed.
2. Click Complete Work Order.
3. Confirm the completion.
4. The Work Order status will change to Completed, and finished goods will be updated in inventory.

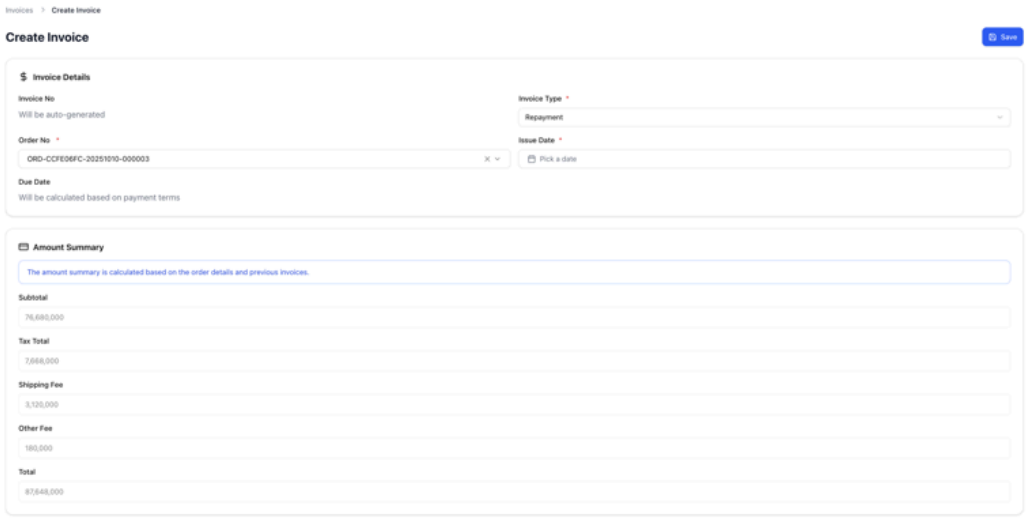
#### Work Order Status Flow

- Not Started → Work Order created but not yet processed.
- Ready to Process → Materials allocated and ready for production.
- In Progress → Routing stages are ongoing.
- Completed → All stages are finished; goods updated in inventory.
- Canceled → Work Order terminated or voided before completion.

---

#### Step 10: Create  Invoice

A  **Invoice** can be created either from the **Order Detail** page or directly from the **Invoice** menu. This allows users to record repayment / down payment transactions linked to specific orders.



#### From Order Detail

1. Navigate to the desired **Order** page.
2. In the **Invoices** section, click **Add Invoice**.
3. Select the **Invoice Type** as *Repayment*.
4. The **Order Number** will be filled in automatically.
5. Click **Save** to create the repayment invoice.

#### From Invoice Page

1. Navigate to **Finance → Invoices**.
2. Click **Create Invoice** on the top-right corner.
3. Choose the **Invoice Type** as *Repayment*.
4. Select or input the **Order Number**.
5. Click **Save** to complete the creation.

---

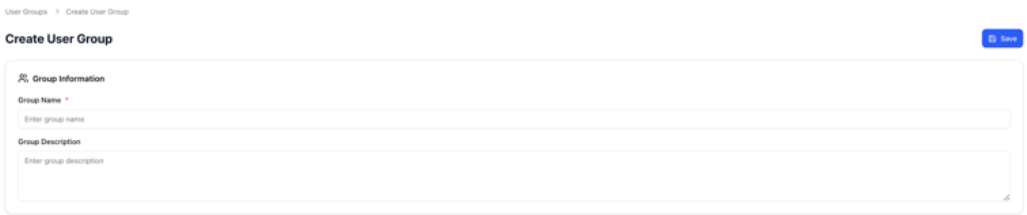
### User Management Setting

The **User Management** module allows administrators to manage user groups, assign permissions, and control access to specific menus and actions within the system.

#### How to View User Group List

1. Navigate to **Administration → User Management → Groups**.
2. The system displays a list of all existing user groups.

#### How to Create User Group



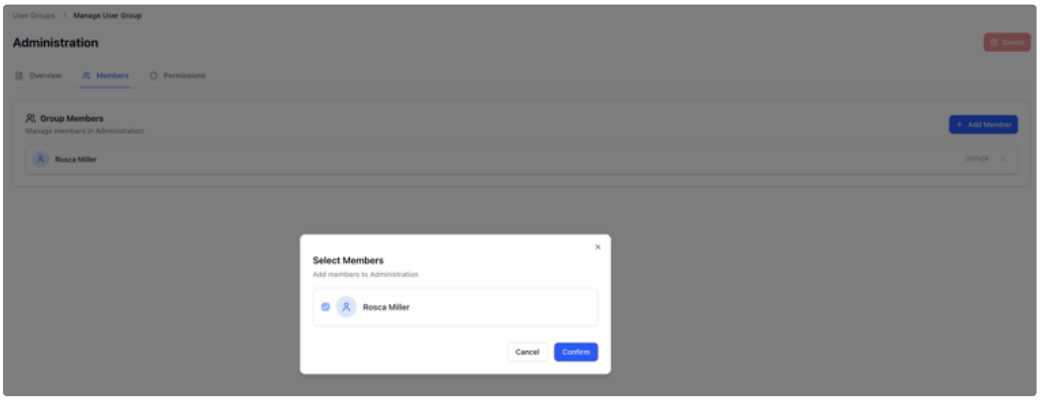

1. Click **Create New Group**.
2. Fill in the following fields:
   - **Group Name** *(mandatory)*
   - **Description** *(optional)*
3. Click **Save** to create the group.

**Notes:**

- The new group appears in the group list.
  If the group name is blank or already exists, the system displays error message.

#### How to Manage Members

### Add Member:

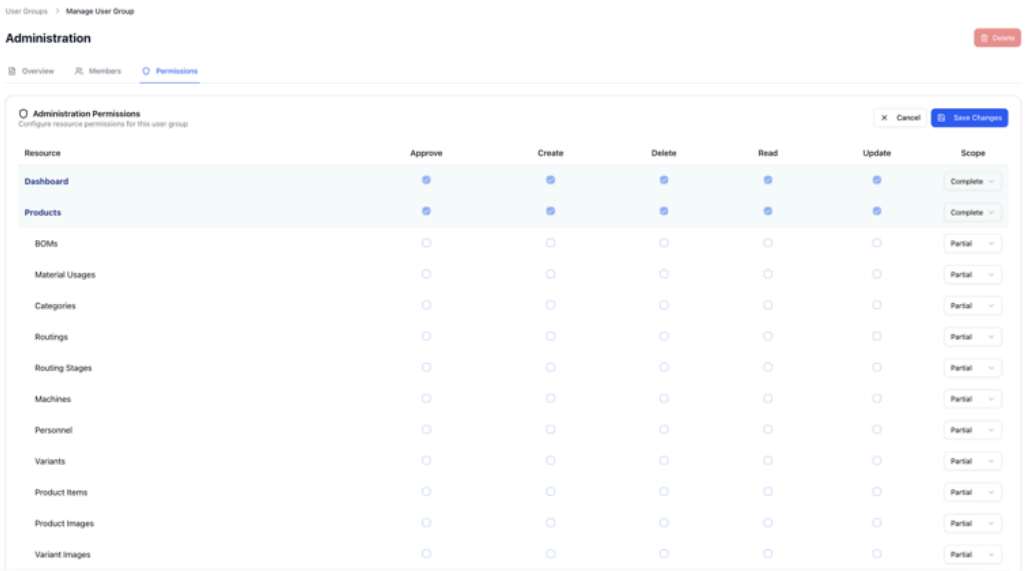



1. Open an existing group.
2. Click **Add Member**.
3. Select the user(s) to be added.
4. Click **Save**.

### Delete Member:

1. From the member list, click **Delete** next to the user.
2. Confirm the removal.

#### How Manage Permissions




1. Open an existing group and navigate to the **Permissions** tab.

2. Select the member whose permissions will be managed.
3. Choose the accessible **Modules** and corresponding **Actions** (Approve, View, Create, Edit, Delete).
4. Click **Save Permissions**.

#### How to Delete Group

1. Select the group you wish to delete.
2. If the group contains members, the system prompts:

   "Deleting this group will remove members' access. Continue?"
3. Confirm the deletion.

**Notes:**
The group is deleted after confirmation, and all related permissions are removed.

---

### FX Management Setting

The **FX Management** module allows administrators to manage currencies, define exchange rates, and control the system's base currency.

#### How to Manage Based Currency

1. Navigate to the **FX Management** section.
2. Review the current **Based Currency** setting.

**Notes:**

- The based currency is currently fixed and **cannot be changed** once it selected through the system.
- Any conversion rates are always calculated relative to this based currency.

#### How to View Currency List

1. Navigate to **Administration → FX Management → Currency List**.
2. The system displays a list of all existing currencies.

#### How to Add New Currency

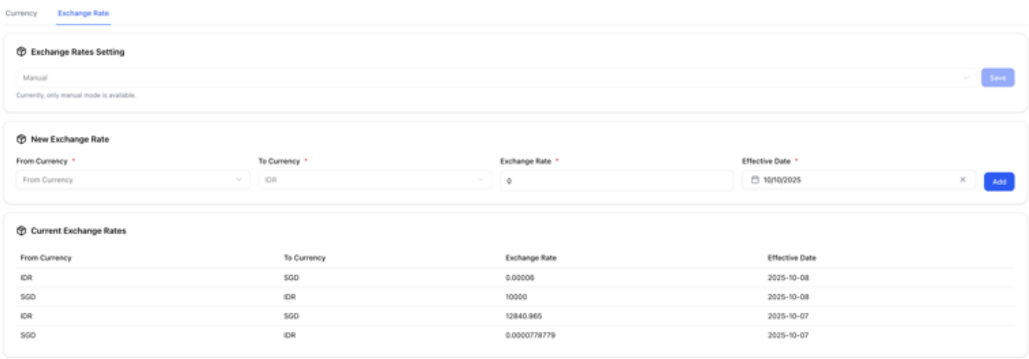
1. Click **Add Currency**.
2. Choose Currency want to add.
3. Click **Save**.

**Notes:**

- The new currency appears in the list immediately after saving.
- If the currency code already exists, the system displays an error message:

  “Currency already exists.”

#### How to Manage Exchange Rates




1. Open the **Exchange Rate** tab.
2. Click **Add New Rate**.
3. Fill in the following fields:
   - **From Currency** – Select the source currency for the transaction (e.g., USD).
   - **To Currency** – Select the target currency for conversion (Default: Base Currency).
   - **Exchange Rate** – Input the conversion rate between the selected currencies (e.g., 15,500 IDR).
   - **Effective Date** – Specify the date when the exchange rate becomes valid (e.g., 10 Oct 2025).
4. Click **Save**.

**Notes:**

- The rate is saved and applied automatically from the specified effective date.
- If the effective date is not entered, the system displays an error message